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BrAINWARS 2024
ROUND 3 SUBMISSION

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Executive Summary: Adopting 'CODE' strategy to optimize cost and efficiency; reviving ZenSoft's profitability issues

BAIN CAPABILITY NETWORK

SITUATION

- Zensoft, a desktop app developer is facing a **decline in profits**
- It **operates via traditional physical servers** leading to high OpEx
- It also grapples with an **aging tech stack** & protracted time-to-market
- It aims to **increase storage capacity to atleast 140TB/mo** on Cloud

MACRO ANALYSIS

- **60%** of all corporate data is stored in the cloud
- **16.3%** Projected CAGR for the global cloud computing market
- **~70%** Organisations opt for hybrid cloud for flexibility, security & latency
- **84%** Reduction in Carbon emissions due to cloud adoption

QUESTION

What are the potential causes for profit decline and where does ZenSoft position among competitors?

Which cloud vendor offers optimal synergy with ZenSoft's constraints for shift from the traditional IT infrastructure?

How can ZenSoft improve its lack of AI capabilities and solve for the aging tech stack?

ANSWER

Cutting EdgeTech Adoption	**Modern Tech Stack (QT Group)**
Optimized Cloud Deployment	**Shift to Hybrid Cloud**
Deep Integration of AI	**AI Capabilities Deployment**
Efficient Team Formulation	**Lean Team Training**

- **<6 Months** Payback Period for cloud migration
- **31.63%** Annual cost savings through AI adoption
- **24.62%** Cost Savings by adopting Modern Stack
- **25%** Cost Savings through Lean Team Training

Notes: 1. Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Gartner (2023), Altantech

Problem Identification & Competitor Analysis: Physical Server Model tends to be the key issue for ZenSoft amidst industry's shift to virtualisation

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Breaking down ZenSoft's costs through a value chain framework…

Profits

Revenues
- Software Sales
- Maintenance & Support Services
- Customisation Services

Costs

Data Infrastructure
- Server and Rental Cost
- Maintenance (electricity, repairs, etc.)
- IT Infra Personnel

Develop-ment
- Technology Stack
- UI/UX Design
- App Developers

Deploy-ment
- Distribution Platform
- Licencing Fees
- CDN Services

Post Delivery
- Testing & Bug Fixes
- Performance Analysis
- SDLC Costs

POTENTIAL ROADBLOCKS IDENTIFIED

Physical Server Cost
78% costlier than virtualisation - based alternatives

High Server Downtime
Gartner reports **downtime costs up to $5,600/min**

Slow Software Update Cycle
Lack of automation slows updates, bug fixes & new launch

…while analysing gaps between industry leaders and ZenSoft

	Operating on On-Prem Physical Servers			*Operating on Cloud Technology/Virtualization*	
Particulars	**ZenSoft**	**TechDesk**		**GenTech**	**SoftGen**
Server Cost	$ 30 mn	$ 33.75 mn[1]	**16%** Reduced Costs	$ 30 mn[1]	$ 20 mn
Storage Cost	$ 2.4 mn	$ 2.7 mn		**N.A.**	**N.A.**
Total Cost	$ 33.9 mn	$ 38.01 mn		$ 36.2 mn	$ 24.1 mn
PCR	**0.884**	**0.887**		**0.827**	**0.828**

High storage costs of on-premise physical servers

Lowered storage costs & PCR reduction from their cloud reliance

Primary Cost Ratio (PCR) = Primary Costs / Total Costs

KEY INSIGHTS

PCR ⬇
PCR is low for cloud-based companies due to reduced **server costs**

~$5 mn
Additional storage costs **for physical servers** compared to cloud

~50% ⬆
More apps launched by **Cloud developers** compared to **primitive**

Notes: 1. Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Gartner (2023), Altantech

Hypothesis Generation: Current problems hint towards 30% opportunity cost due to persistence with traditional and outdated technologies

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HYPOTHESIS

1 — **Physical servers pose high cost, operational inefficiencies which can be solved via cloud**

2 — **Primitive stack incurs high storage cost, low uptime due to inadequate modernization**

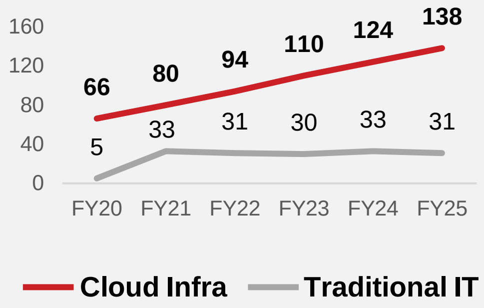
3 — **Lack of AI slows bug fixes, updates and leads to rise in application development cost**

4 — **High IT staff costs managing physical servers necessitates lean teams, trained internally**

SUPPORTIVE EVIDENCE

1

Growth in Cloud Revenue (in Bn)



2

$1.8 bn

Annual Productivity lost due to outdated technology (Forbes report)

~ 70%

Of IT Expenditure is spent on maintaining primitive stack in the US

3

~ 75%

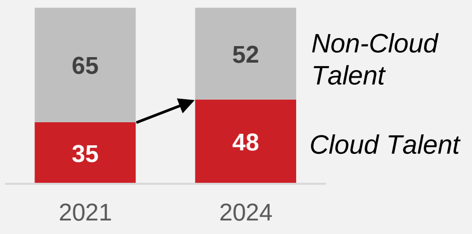
Potential ROI increase with Generative AI

65%

Companies get their AI capabilities through cloud-based software

4

Internal Personnel Training



To make up for cloud skill gaps, companies internally reskill IT talent

IMPLICATIONS DRAWN (PAIN POINTS)

1 — ZenSoft's **$2.5M** on physical servers indicates infrastructure inflexibility & prolonged time-to-market cycle. Without cloud adoption, **potential 54.96% savings** are foregone.

2 — ZenSoft's storage inadequacy and downtime risks suggest **potentially forfeited cost savings of 24.62%** by not transitioning to a pay per view, cloud based modern stack.

3 — ZenSoft's contract loss from lacking **CI/CD** via AI impacts storage, updates, and automation, posing **$40,000 opportunity cost** for faster launches and deployments.

4 — ZenSoft spends $1M on IT staff for physical servers, optimizable through lean virtualization training, currently **18% higher operational costs** when continued unchanged.

IMPACT SIZE

1 — LOW / HIGH

2 — LOW / HIGH

3 — LOW / HIGH

4 — LOW / HIGH

Source: Gartner (2024), McKinsey Tech Report (2023), Morgan Stanley (2023), Nutanix (2023), Forbes (2022)

Cloud Vendor Selection: Digital Ocean ranks as the optimal cloud service vendor based of EVI Index and problem-based 4 KPIs

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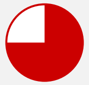
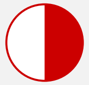
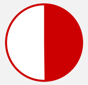
CHOICE OF CLOUD VENDORS

KPIs	DigitalOcean	UpCloud	rackspace the open cloud company	DIGITAL REALTY
Product Offerings	Droplets, Load Balancer Kubernetes, PaaS, Volumes	Compute, Custom ISOs Networking, MaxIOPS, Backups, Monitoring	Hybrid & Multi Cloud, Managed Hosting and Database Services	Data Centres, Managed Services, Interconnection, Hyperscale, Data Hub
Core Competency	**Developer-friendly**, simple & affordable infrastructure	**Block Storage**, High Performance Computing	Comprehensively managed cloud services	Secure, scalable **data center infrastructure**
Synergy with Zensoft				
Enhanced Value Index* (Lower is better)	**0.2333**	**1.1552**	**1.2667**	**0.2440**

Vendor Alternatives →

Favorable-ness of the factor

UF
ME
SE
CE

UF = Update Frequency, ME = Maintenance Efficiency, SE = Service Excellence, CE = Cost Effectiveness
Darker the shade, more favorable the impact of the factor by the vendor (backed by data in the case).

Enhanced Value Index (EVI)*

=

Total Service Cost / **(Storage Capacity) x (Update Cycle Multiplier)**

+

Security Penalty

+

Storage Insufficiency Score

A lower EVI signifies better cost efficiency and service quality

EVI amalgamates **financial & operational factors**, empowering ZenSoft to objectively evaluate cloud vendors' cost effectiveness and efficiency via an all-inclusive score

Refer to Appendix for Deep Dive

ZenSoft's preferred partner, **Digital Ocean**, emerges from a scoring model that **factors in critical pain points:** costs, software update cycles, and service excellence.

Notes: 2. Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Company websites, Forrester(2022)

ZenSoft x Digital Ocean: With ROI of 186% and a payback within 6 months, onboarding Digital Ocean offers potential product synergies

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DigitalOcean's solutions for developers synergise best with Zensoft

ESTIMATED COST ~ $19000/ MONTH

DOCKER | Node.js | UBUNTU | KUBERNETES | POSTGRES SQL

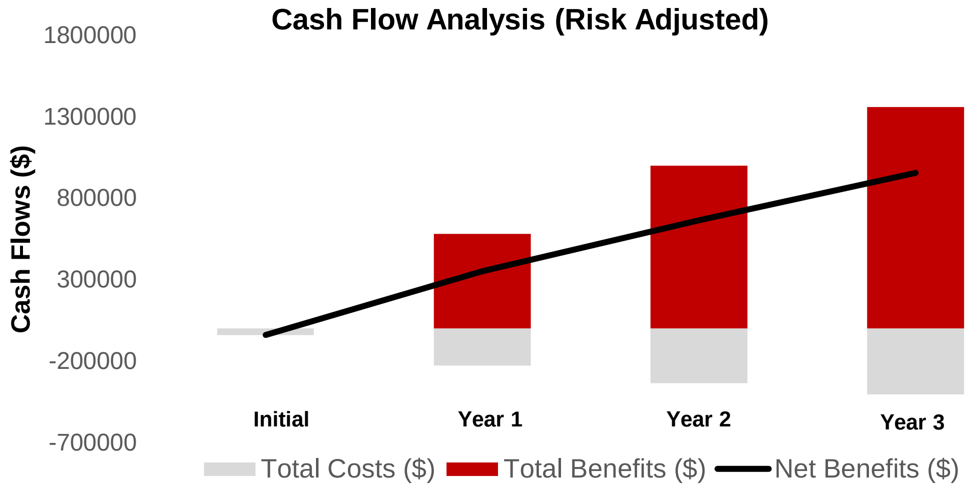
The above offerings, combined with Backups, Storage, Networking and Uptime Solutions via Developer Tools provide a package for ZenSoft

Drawing Parallel Synergies with Clients in the Tech Space…

Validin | teraone | Clientify

- **Comprehensive One Stop Public Cloud Solution**
- **Security and Reliability for Global Operations**
- **Increased Productivity valued at $300,000**
- **Scalable Virtual Machines with > 50% savings**

Cost-Benefit Analysis highlights Key Financials…



Net Benefits show increasing trend due to substantial rise in benefits, despite rising cost. This stands in line with industry trends.

… Which indicate Potential Gain for ZenSoft

~$2Mn	**~186%**	**< 6 Months**
Cumulative Net Benefits spread across 3 years	ROI of Developers (Clients of Digital Ocean)	Payback Period for the Partnership (Forrester)

Notes: 1. Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Economics of Digital Ocean – Forrester(2022), Digital Ocean website

Further Strategic Considerations: Aligning with industry's trends toward multi-hybrid cloud by analysing Vmware and VPC's private cloud

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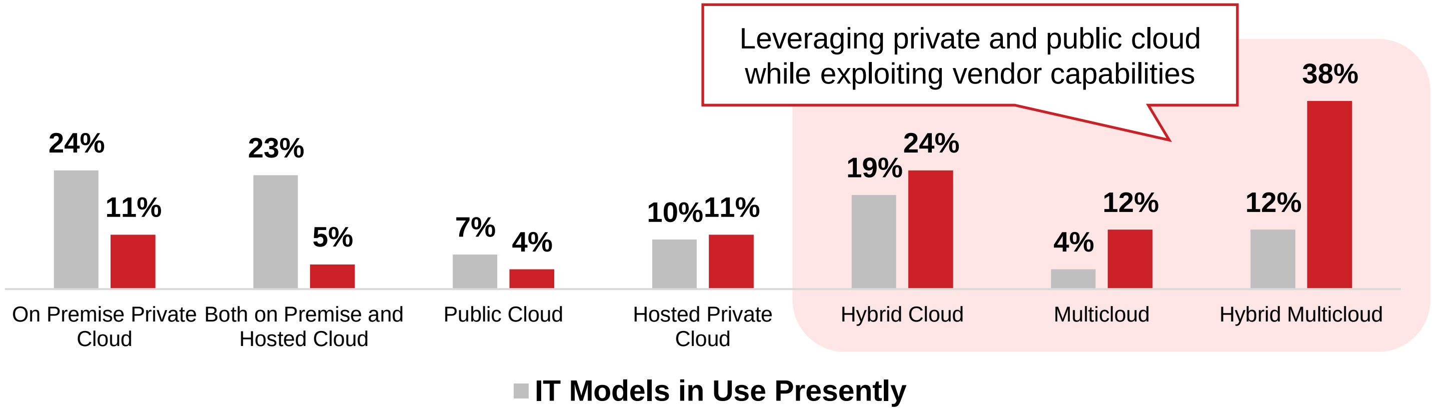
The market has seen a shift towards hybrid & multi cloud arrangements...

Deployment Models currently in use and planned for use in the next 1-3 years (%)



Why choose the hybrid cloud deployment over public or private cloud?

FACTORS	PUBLIC	PRIVATE	HYBRID
Initial Setup	Easy	Complex	**Complex**
Scalability and Flexibility	High	Low	**High**
Cost Comparison	Inexpensive	Costly	**Cost-effective**
Reliability	Low	High	**High**
Data Security and Privacy	Low	High	**High**

Hybrid cloud allows users to circumvent the lack of security and reliability while maintaining scalability and cost efficiency

…and ZenSoft can do so in either of the two ways

1. SETUP HYBRID CLOUD BY ONBOARDING VMWARE

Utilizing the existing On-Prem Infra to onboard VMVare as a private cloud vendor, specializing in app-dev companies.

PROS		CONS	
	Increased Data-Safety Multi-Cloud approach, reducing over-reliance		More Expensive Integration Challenges and Complexities

2. LEVERAGE VIRTUAL PRIVATE CLOUD BY DIGITALOCEAN

VPC provisions logically isolated sections of a public cloud in order to provide a virtual private environment.

PROS		CONS	
	Cheaper Alternative Smoother and easier integration		Higher Data Breach risk Vendor Lock-in, being over-reliant

VMware proves to be a viable option for ZenSoft which can utilize its existing servers and would require no additional capex. It also results in positive synergies owing to no vendor lock-in and specialisation in app development.

Notes: 2. Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Gartner (2024), McKinsey Tech Report (2023), Morgan Stanley (2023), Azure Website

Modern Tech Stack: QT Group serves as ZenSoft's ideal partner in transitioning to modern stack fixing system downtime and time-to-market

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Letting go of primitive stack in favour of modern technology…

WHY CHOOSE MODERN STACK?

- ✓ Modern stacks **reduce long-term operational costs by up to 30%,** providing substantial savings compared to the heavy maintenance expenses of outdated systems.
- ✓ Upgrading to a modern stack can offer up to **80% in energy cost savings**, enhancing operational efficiency with the latest energy-efficient technologies.
- ✓ With over 35% cost savings on cloud services and rapid service integration, modern stacks **ensure scalable and agile business operations.**

OUTDATED VS MODERN TECH STACK

ECONOMIC FACTOR	OUTDATED STACK	MODERN STACK
Initial Investment	Lower Upfront Costs	Higher Upfront Costs
Maintenance Cost	Higher	Lower
Scalability & Flexibility	Costly & Cumbersome	Cost Effective & Flexible
Operational Efficiency	Lower	Higher
System Downtime	More Frequent	Less Frequent

…implemented via shifting to QT Group's modern architecture

SELECTION OF THE PROVIDER

Provider	Platform Support	Performance	Developer Community
WPF (Windows Presentation Foundation)	Windows	High	Large
UWP (Universal Windows Platform)	Windows	Variable	Large
COCOA	Mac OS	High	Large (for Apple Developers)
ELECTRON	Cross Platform	High	Large
Qt Group	Cross Platform	High	Large

Supports desktop, mobile embedded systems, provides high performance and access to native OS-dependent libraries. It **fits the requirements of ZenSoft as a Desktop Application Development Company.**

Notes: 2. Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Gartner (2024), McKinsey Tech Report (2023),QT Group Website, GitHub

Streamlined AI Integration: Optimizing Cloud Server Efficiency with Digital Ocean and CTO.ai, achieving up to 50% KPI enhancement

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CI/CD Integration for ZenSoft's AI Capabilities…

What is CI/CD Pipeline for AI?

Continuous Integration & Continuous Deployment

- ✓ **Automating** the build, testing of code using SaaS Build Device (long term) or on-premise build server (short term).
- ✓ Keeps **software development cycle** on track by ensuring that errors or bugs are fixed as spotted.
- ✓ Deploys code with **zero touch automation.**

… Which intend to solve ZenSoft's Challenges

High

Automated Deployment: Speeds up software development and new product launches.

Smart Code Reviews: Lowers software development costs and reduces bug counts.

Predictive Analysis & Intelligent Automation: Quick error detection and software development.

Self-Healing Builds: Reduction in downtime and build failure resolution time.

Low

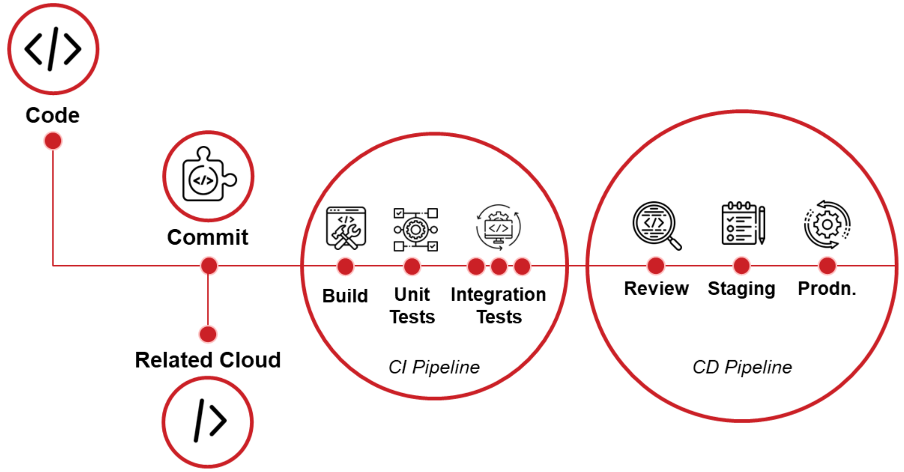
Source: Digital Ocean website, CTO.ai Website, Dreamhost

…implemented through CTO.ai + Digital Ocean

CTO.ai *Digital Ocean's trusted partner for AI implementation for DevOps as a Service*

Has worked with existing Digital Ocean clients such has Validin, Snipitz, and Jiji to provide satisfactory results.

How is the Pipeline Automated?



CI/CD Automation involves putting code changes in one place, testing them automatically, and then putting them into use

KEY IMPACT

~90%
Reduction in Deployment Time

~50%
Setup Time Optimisation

~75%
Data Efficiency Enhancement

~30%
Reduced Infrastructure Costs

~40%
Faster Mean Time to Resolution

(As per CTO.ai impact reports)

Lean Team Training: With over 70% leaders preferring hybrid cloud, formation of lean team with internal training projects potential savings of 25%

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As we move to a cloud infrastructure, personnel modification according to the needs of the company is crucial…

As cloud migration for ZenSoft takes place, instead of reducing staff, creation of new roles with new skills is required…

…for this two cases may arise: laying off existing employees to hire new talent or Leaner IT Staff is internally trained for cloud capabilities

CASE 1: SUPPOSE 40% OF THE CURRENT IT STAFF IS LAID-OFF AND CLOUD EXPERTS ARE HIRED

Layoffs due to cloud migration and hiring cloud professionals for the same results in **additional expenses of ~$ 0.2 Mn as a result of severance costs. Also causes reputational damage.**

CASE 2: LEANER IT STAFF UNDERGOES CLOUD TRAINING AND REQUIRED CLOUD EXPERTS ARE HIRED

Training existing IT staff and hiring cloud professionals proves to be a more **cost-effective option and reduced reputational damage** which will have a favorable impact on the ESG performance.

~25% **Potential Cost Savings arising from Case 2**

…which can be achieved through a streamlined process

Analye Additional Workload

Identify tasks associated with managing cloud services (e.g., supplier management, monitoring).

Identify potential Skill Gap

Compare existing staff skills with those required for cloud mgmnt. (e.g., service-level management, customer expectations).

Assess Current Cabilities

Analyze the service desk's ability to handle cloud-related issues and incidents. Consider potential upgrades or training needs.

Form Teams and Team Leads for Migration

Form a Cloud Initiative Team to spearhead cloud migration planning and implementation.

Plan Training and Development Modules

Outline training programs or recruitment strategies to address skill gaps. Factor in additional migration costs (staff, tools, training) for transitioning to cloud management.

Notes: 2. Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Gartner(2024), McKinsey Tech Report (2023), Deloitte(2021)

ESG Impact Calculation: Evaluating ZenSoft's ESG impact following its investment concerning GHG emissions, labor practices, and data privacy.

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Implementation of solution has helped ZenSoft improve its ESG score…

ESG Score for ZensSoft prior to cloud adoption = **1.625/3****

ESG PILLAR	ZENSOFT'S ISSUE	POINT	WEIGHT ALLOTTED	WEIGHTED SCORE
ENVIRONMENTAL			**0.3**	
	Energy Management	2	0.075	0.15
	Greenhouse Gas Emissions	3	0.15	0.45
	Waste Management	1	0.075	0.075
SOCIAL			**0.35**	
	Labour Management	3	0.1375	0.4125
	Diversity, Equity & Inclusion	2	0.1375	0.275
	Community Engagement	2	0.075	0.15
GOVERNANCE			**0.35**	
	Data Privacy	3	0.15	0.45
	Corporate Transparency & Disclosure	2	0.1	0.2
	Management Structure & Compensation	2	0.1	0.2
TOTAL SCORE			78.75%	2.3625/3

PERFORMANCE: Strong | Average | Weak

Methodology Adopted

A Determine relevant material issues by comparing each material sources to ZenSoft's Characteristics

B Create a scale and define clear and meaningful criteria that reflects ESG performance.

1	2	3
No policies, goals, strategies, performance outcomes/ disclosed data	Partial Coverage policies, goals, strategies, performance outcomes/disclosed data	Clearly stated policies, goals, strategies, performance outcomes/disclosed data

C Weight Variables by their level of relative importance in the industry and to the stakeholders

30% Governance Issues	35% Social Issues	35% Environmental Issues

Zensoft's ESG performance **post implementation of ESG initiatives is strong (in this case cloud migration)**, the total score 2.362/3 indicates investing in suggested use cases will be lucrative for the company.

Notes: ** Refer to Appendix for Calculation Assumptions
Source: Team Analysis, Refinitiv website

Risk-Mitigation Analysis: Recommended solutions pose key risks for ZenSoft which need to be mitigated to enhance long term profitability

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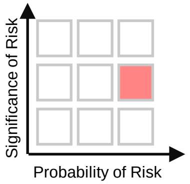
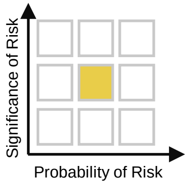
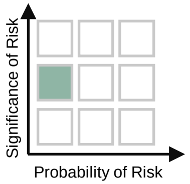
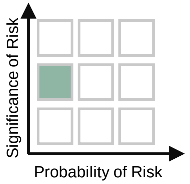
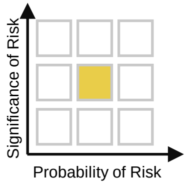
SOLUTIONS PROPOSED	KEY RISKS	IMPACT OF RISK	POTENTIAL MITIGATION
Public Cloud Deployment via Digital Ocean	Excessive reliance on single cloud provider **(Vendor Lock-in)** and concerns regarding **security breaches** during data transfer.		Considering a **hybrid cloud strategy,** blending public & private providers with implementation of **data tokenization** to minimize breach risks.
Hybrid Cloud Migration: Digital Ocean & VMware	Increased **operational complexity** and management overhead along with **higher upfront investment** and **poor encryption.**		Implementing industry standards and **APIs** for smooth integration between public and private clouds, also ensuring **regulatory compliance.**
Integration of Gen AI Capabilities into Cloud	Possibility of **automated deployment errors**, **security vulnerabilities** and low **accuracy**, resulting in **service disruptions.**		Continuous monitoring, rigorous testing with **CI/CD** and a **trusted partner**, mitigate security and deployment risks, aiding **high accuracy**.
Lean Team Formation and Staff Training	**Diversion of resources** from critical business activities and **loss of trained employees** to other opportunities, disrupting current projects.		**Retention strategies**, **succession planning** ensures employee retention, mitigating turnover impact and ensuring smooth operations.
Adopting Modern Tech Stack with QT Group	**Compatibility issues, downtime** and unexpected costs, including licensing fees and training expense pose significant potential risk.		Minimize operational disruption through in- depth planning, **phased rollouts**, and **A/B pilot tests**, while also conducting **cost-benefit analysis.**

Axes of each risk matrix: Significance of Risk (vertical), Probability of Risk (horizontal)

Low Risk | **Medium Risk** | **High Risk**

Source: Team Analysis

Implementation Timeline: Proposing a feasible timeline to be implemented to aid ZenSoft's cost efficiency and profitability

BAIN CAPABILITY NETWORK

2024 | 2025 | 2026+

June | Oct | Jan | Mar | June | Oct | Jan | March

MODERNIZATION OF OUTDATED TECH STACK
- Modernisation Process carried out to lower Downtime
- Streamlining Processes to further achieve Increased efficiency
- R&D carried out to integrate modern elements and prevent outdation
- Identifying prospective areas for further downtime reduction.

PUBLIC CLOUD MIGRATION VIA DIGITALOCEAN
- Assessing operational fluency and smooth integration through pilot & A/B testing
- Securely transitioning data from physical servers to public cloud
- Exploring multi-cloud options to ensure diversification

HYBRID-CLOUD ADOPTION
- Identifying Private vendors and explore Virtual Private Cloud option under DigitalOcean
- Assessing operational fluency and smooth integration through pilot testing
- Gradual switch from remaining physical servers to hyrbrid cloud

AI INTEGRATION
- AI Integration planning and setup environment
- Integrate Tools from the onboarded partner into DigitalOcean
- Test Integration and configuring triggers to find errors
- Refining workflows to ensure optimize all integrated AI Capabilities

EMPLOYEE TRAINING
- On the Job training: Modern Stack and Moving towards Cloud
- Talent Retention strategies and success planning aiming to prevent brain-drain
- Gradual Up-skilling related to modern technologies such as Edge Computing

Phase of Operation | Must-Win Battles | Feedback | Contingent on Prior Successes

Source: Team Analysis

APPENDIX

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APPENDIX (1/15): Approach and Assumptions

1. **Nature of the Case:** Profitability and Cost Optimization (Company has declining profit due to inefficiencies in operations & technological capabilities).

2. **Approach used:**
   - Analysing ZenSoft's cost structure to identify potential roadblocks and cost reductions through value chain cum profitability framework.
   - Computation of Primary Cost Ratio to benchmark competitors based on server costs.
   - Agile transition to modern stack eventually reducing untenable operational costs and improving current market positioning.
   - Swift adoption of hybrid cloud model to gather benefits of both private and public cloud, ensuring data security and optimised cost.
   - Implementation of Gen AI technologies to automate continuous integration/continuous deployment (CI/CD) and boost software development cycle.
   - Identifying internal training methodologies and weighing them over staff-cuts, ultimately leading to significant cost savings

3. **Reasons for selecting the profitability cum value chain framework:**
   - More insightful and holistic in nature as compared to other frameworks.
   - Specific Identification of problems due to stage-wise breakdown, enabling assessment of cost drivers.
   - Operations oriented (conducive in analysis of specific cost structure and evaluate potential cost savings under each approach).
   - More fitting to the situation

4. **Assumptions made:**
   - ZenSoft has its operations not restricted to any specific country but are carried out worldwide.
   - Desktop application services provided by the company are not limited to a particular industry / sector.
   - Costs projected and savings shown are with respect to a desired storage capacity of 150 TB.
   - There are no financial and geographical constraints with regards to migration to cloud servers and adoption of Gen AI capabilities.

APPENDIX (2/15): Peer Comparison; Physical vs Cloud – Cost Structure

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Companies	On-Premise Physical Servers		ASSUMPTIONS
	Particulars	**XXX ($)**	
ZENSOFT	Server Cost (pa)	30000000	
	App Development Costs	500000	
	Storage Cost of Servers	2400000	
	Staffing costs	1000000	
TECHDESK	**Server Cost (pa) ~**	**33750000**	*Calculated using the same ratio between server costs incurred by Zensoft & Techdesk*
	App Development Costs ~	**562500**	
	Storage Cost of Servers	2700000	
	Staffing costs	1000000	
	TOTAL COSTS	**71912500**	

Companies	Cloud Server		ASSUMPTIONS
	Particulars	**XXX ($)**	
GENTECH	**Server Cost (pa)**	**30000000**	*Based on industry averages, Gentech, as the leader, will likely bear server costs comparable to Softgen's*
	App Development Costs	350000	
	IT Handling Costs	**900000**	*IT handling costs reduction incorporates leaner team due to absence of storage costs*
	Other Transmission costs	**5000000**	*Additional transmission costs encompass data transfer and other expenses related to server usage, based on industry norms*
SOFTGEN	Server Cost (pa)	20000000	
	App Development Costs	400000	
	IT Handling Costs	**750000**	*IT handling costs reduction incorporates leaner team due to absence of storage costs*
	Other Transmission costs	**3000000**	*Additional transmission costs encompass data transfer and other expenses related to server usage, based on industry norms*
	TOTAL COSTS	**60400000**	*Cloud servers notably cut expenses compared to physical ones, largely due to reduced storage costs*

APPENDIX (3/15): Peer Comparison; Calculation of Primary Cost Ratio

COMPANIES	PRIMARY COSTS	TOTAL COSTS	PRIMARY COST RATIO
ZENSOFT	30000000	33900000	0.884955752
TECHDESK	33750000	38012500	0.887865834
GENTECH	**30000000**	**36250000**	**0.827586207**
SOFTGEN	20000000	24150000	0.82815735

ASSUMPTION:

Primary Costs involve only Server costs incurred by all 4 companies, thereby negating the effect of secondary and tertiary costs and focusing on the key aspect of server cost comparison between physical and cloud servers.

KEY INSIGHT:

Companies operating on cloud servers i.e. Gentech and SoftGen appear to be saving close to 16% costs in contrast with physical servers wherein the major costs incurred revolve around storage costs of physical servers and the extensive amount spent on It handling staff.

APPENDIX (4/15): Cost Savings – Hybrid Cloud Vs On-Premise Servers

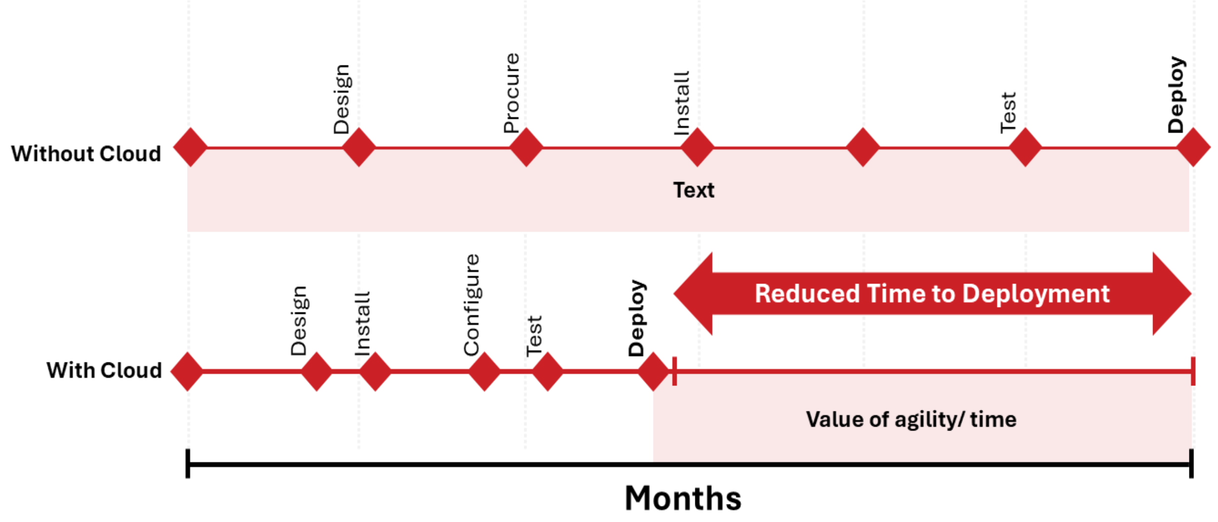
BAIN CAPABILITY NETWORK

Cost Savings - Hybrid Vs On-Premise servers					
Cost Component	**On-Premise Server Cost**	**DigitalOcean Cloud Cost**	**Hybrid Model Cost**	**Net Savings**	**Assumptions/Sources**
CAPEX (Capital Expenditure)					
Data Center Infrastructure	$ 1,00,000.00	$ -	$ 30,000.00	$ 70,000.00	Cost of physical servers, networking hardware, etc.
Server Provisioning	$ 25,000.00	$ -	$ 7,500.00	$ 17,500.00	Man-hours for server setup, OS installation
OPEX (Operational Expenditure)					
Compute Instances (VMs)	$ 50,000.00	$ 35,000.00	$ 40,500.00	$ 9,500.00	Virtual Machine instances, based on DigitalOcean Droplets
Block Storage	$ 20,000.00	$ 14,000.00	$ 15,800.00	$ 4,200.00	SSD-based block storage, DigitalOcean Volumes
Object Storage	$ 5,000.00	$ 3,500.00	$ 4,100.00	$ 900.00	S3-compatible storage, DigitalOcean Spaces
Data Transfer Outbound	$ 10,000.00	$ 7,000.00	$ 8,400.00	$ 1,600.00	Based on bandwidth usage and DigitalOcean Transfer Pricing
Load Balancer Services	$ 3,000.00	$ 2,100.00	$ 2,430.00	$ 570.00	DigitalOcean Load Balancer costs
Managed Database Services	$ 15,000.00	$ 10,500.00	$ 12,150.00	$ 2,850.00	Managed services such as DigitalOcean Managed Databases
Container Orchestration	$ 5,000.00	$ 3,500.00	$ 4,100.00	$ 900.00	Kubernetes orchestration, DigitalOcean Kubernetes service
Network Security	$ 8,000.00	$ 2,800.00	$ 4,240.00	$ 3,760.00	Firewalls, VPNs, and other network security solutions
System Administration	$ 30,000.00	$ 21,000.00	$ 24,300.00	$ 5,700.00	SysAdmin costs for managing hybrid infrastructure
DevOps Automation Tools	$ 7,000.00	$ 4,900.00	$ 5,670.00	$ 1,330.00	CI/CD pipeline tools, infrastructure as code services
Compliance & Security	$ 25,000.00	$ 17,500.00	$ 20,250.00	$ 4,750.00	Regulatory compliance, audits, and security monitoring
Disaster Recovery	$ 15,000.00	$ 10,500.00	$ 12,150.00	$ 2,850.00	DRaaS (Disaster Recovery as a Service)
Staff Training	$ 10,000.00	$ 7,000.00	$ 8,100.00	$ 1,900.00	Training for cloud and hybrid cloud operations
Total Annual OPEX	**$ 2,03,000.00**	**$ 1,39,300.00**	**$ 1,62,190.00**	**$ 40,810.00**	
Initial CAPEX + Annual OPEX	**$ 3,28,000.00**	**$ 1,39,300.00**	**$ 1,99,690.00**	**$ 1,28,310.00**	

Cost Savings(%)	**39.12%**

APPENDIX (5/15): Impact Generation and Process of Cloud Migration along with an additional revenue stream through existing physical servers

Shifting to cloud creates impact, reducing time to deployment & become more efficient…



Process of implementing the migration from on premise physical servers to a hybrid cloud

1. **Assess Current infrastructure and Workload**
2. **Define Objectives and Requirements**
3. **Select the right cloud providers**
4. **Design the hybrid cloud architecture**
5. **Implement hybrid cloud infrastructure**
6. **Ensure data security and compliance**
7. **Testing and Optimization**
8. **Monitoring and Management**

Usage of Remaining Physical Servers

Revenue Generation Potential

- ✓ Renting out or collocating additional physical servers.
- ✓ Additional revenue of up to **15%** of IT Budget.
- ✓ Global colocation market growing at **12.1%** CAGR.

Disaster Recovery

- ✓ With an average downtime cost of **$5600**/minute, 87% of the organizations experience disruptions in operations.
- ✓ Reduction of financial impact of downtime and mitigate risks.
- ✓ Hybrid disaster recovery via cloud and on-premise servers witness **27% less** downtime compared to solely relying on traditional methods.

APPENDIX (6/15): Cloud Provider Scoring Model based on 4 KPIs

Vendor	UF Rank	ME Rank	SE Rank	CE Rank	Total Score	Assumptions and Sources
Rackspace	3	3	4	3	13	*Assumed worse service record due to litigation issues; no recent awards noted. Sources: Industry reports, legal news.*
DigitalOcean	1	2	1	2	6	*Frequent updates and awarded for client service. Competitive pricing and additional storage. Sources: DigitalOcean's official service update logs, Cloud Spectator 2021 Report.*
UpCloud	4	1	1	1	7	*Best maintenance cost ratio, no setup cost, and high client service rating. Sources: UpCloud's service agreements, industry service awards.*
DigitalReality	2	4	4	4	14	*Higher costs noted and less flexibility in services. No recent awards or recognition for client service. Sources: Digital Reality's public pricing updates, industry comparison studies.*

KPI Explaination

Update Frequency (UF):

Assumption: More frequent updates indicate better responsiveness to technology changes and security.

Source: Vendor's official documentation and update logs.

Maintenance Efficiency (ME):

Assumption: Lower maintenance costs as a percentage of annual cost indicate better operational efficiency.

Source: Financial data from vendor reports and industry average benchmarks.

Service Excellence (SE):

Assumption: Awards for client service are indicators of superior customer support and satisfaction.

Source: Industry awards and customer service ratings from independent review sites.

Cost Effectiveness (CE):

Assumption: Evaluating overall value for money in terms of storage per dollar and considering setup costs as a significant initial investment factor.

Source: Comprehensive price-to-feature comparison reports and vendor pricing pages.

Total Score has been devised based on key pain points suffered by ZenSoft, which ranks synergies with potential cloud providers (Lower the score, better the partner):

1. **Digital Ocean**
2. **UpCloud**
3. **Rackspace**
4. **Digital Reality**

APPENDIX (7/15): Enhanced Value Index (EVI) Calculations and Rationale

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Enhanced Value Index(EVI) Calculations										
Vendor	Annual Cost(M)	Maintenance Cost(M)	Setup/Training Cost(M)	Total Service Cost(M)	Storage Capacity(TB)	Update Cycle Days	Update Cycle Multiplier	Storage Insufficiency Score	Security Penalty	EVI Index
DigitalOcean	$ 21.60	$ 0.15	$ 0.80	$ 22.55	145	15	2.67	0	0	0.41
Rackspace	$ 18.00	$ 0.24	$ 0.50	$ 18.74	140	20	1.50	0	1	1.20
UpCloud	$ 20.00	$ 0.18	$ -	$ 20.18	130	10	3.00	1	0	1.47
Digital Reality	$ 25.00	$ 0.21	$ -	$ 25.21	155	15	2.00	0	0	0.33

EVI Formulas

$$EVI = \frac{TSC}{\text{Storage Capacity} \times \text{Update Cycle Multiplier}} + \begin{matrix}\text{Security Penalty} \\ \text{Storage Insufficiency Score}\end{matrix}$$

Assumptions and Sources

Lower EVI scores indicate better cost efficiency and service quality, positioning DigitalOcean as the best and Rackspace as the least attractive option.

The calculations assume specific weights for costs and incorporate a penalty for Rackspace's security issues to

Annual Cost is the total annual expenditure.

Maintenance Cost is weighted normally.

Setup/Training Cost is weighted heavier for Rackspace due to security issues.

Update Weight = 40/Update Cycle Days **to favour short cycles**

Security Penalty applies only to Rackspace, penalizing for past security issues.

Security Penalty: Rackspace is penalized by adding 1 units to their EVI score due to past security issues.

Setup/Training Cost Weight: For Rackspace, setup/training costs are doubled in the formula to reflect higher risk and potential future costs due to security enhancements.

Maintenance and Setup Costs: Assumed to be less impactful for DigitalOcean given its efficient service, hence included linearly.

Storage Insufficiency Score: UpCloud has an added factor of 1 units due to it's insufficiency to provide storage capacity

Lower EVI scores indicate better cost efficiency and service quality, positioning Digital Ocean as the best and Rackspace as the least attractive option.

APPENDIX (8/15): Selecting Digital Ocean as the ideal cloud provider

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On-Premise Vs Cloud Comparison					
Cost Category	**Sub-Categories**	**On-Premise Estimated Costs**	**DigitalOcean Costs**	**% Savings/Cost Impact**	**Assumptions & Sources**
Capital Expenditure (CapEx)	Hardware Purchase	$50,000	$0	100% Savings	*Avoidance of hardware purchase. Source: Techopedia*
	Depreciation	$10,000/year	$0/year	100% Savings	*Depreciation over 5 years. Source: Investopedia*
Operational Expenditure (OpEx)	Power, Cooling, Space	$24,000/year	Included	100% Savings	*Operational costs rolled into cloud service fee. Source: Data Center Knowledge*
	Server Maintenance	$5,000/year	$150,000/year	2900% Increase	*Higher maintenance due to comprehensive service. Source: Gartner*
Personnel Costs	IT Staff for Maintenance	$1,000,000/year	$700,000/year	30% Savings	*Reduction in IT staff due to managed services. Source: Glassdoor*
	Training for Cloud Transition	-	$800,000 (one-time)	N/A	*One-time cost for staff training. Source: Forbes*
Service Costs	Annual Cloud Services	-	$21.75M/year	N/A	*Includes storage, security, updates. Source: DigitalOcean Pricing*
Total Direct Costs	Annual Operating Costs	$1,089,000/year	$22.65M/year	-1996% Increase	*Sum of annual costs including service fees.*
Indirect Benefits & Cost Impact	Scalability and Flexibility	Fixed costs	Variable costs	Significant flexibility	*Scalability as per demand reduces waste. Source: IBM Cloud Learn*
	Speed to Market	Slower	Faster	Speed improvement	*Cloud infrastructure allows quicker deployments. Source: Harvard Business Review*
	Opportunity Costs	Higher	Reduced	Reduced missed opportunities	*Cloud agility avoids delays in opportunities. Source: The Economist*

Competitor Analysis				
Factor	**Rackspace**	**DigitalOcean**	**UpCloud**	**Digital Reality**
Storage Capacity (TB/mo)	140	145	130	155 (updated)
Annual Pricing	$18M	$21.6M	$20M	>$25M (old price)
Annual Pricing Per TB	$0.13	$0.15	$0.15	Not available
Annual Maintenance Cost	$240K	$150K	$180K	$216K
Software Update Cycle	20 days	15 days	10 days	15 days
Maintenance Cycle	25 days	30 days	20 days	30 days
Setup/Training Cost	$0.5M	$0.8M	-	-
Client Service Award	-	Yes	Yes	-
Litigation/Security Issue	Yes	-	-	-
Training Period	3 weeks	3 weeks	-	-
Cost Effectiveness*	-	More cost-effective employees	More cost-effective employees	More cost-effective employees

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APPENDIX (9/15): Cost Benefit Analysis – Digital Ocean as a cloud provider

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Cumulative Cost Particulars						
Cost Categories from DigitalOcean Report	**Initial**	**Year 1**	**Year 2**	**Year 3**	**Three-Year Total**	**Three-Year Present Value**
Cost of DigitalOcean Workload	$ -	$ 2,17,930.00	$ 3,22,831.00	$ 3,91,822.00	$ 9,32,583.00	$ 7,59,302.00
Implementation and Management Cost (risk-adjusted)	$ 34,650.00	$ 5,775.00	$ 6,600.00	$ 7,425.00	$ 54,450.00	$ 50,933.00
Training Cost (risk-adjusted)	$ 5,775.00	$ 4,620.00	$ 5,280.00	$ 5,940.00	$ 21,615.00	$ 18,801.00
Total Annual Costs(Risks-adjusted)	**$ 40,425.00**	**$ 2,28,325.00**	**$ 3,34,711.00**	**$ 4,05,187.00**	**$ 10,08,648.00**	**$ 8,29,036.00**
Cumulative Benefit Particulars						
Benefit Categories from DigitalOcean Report	**Initial**	**Year 1**	**Year 2**	**Year 3**	**Three-Year Total**	**Three-Year Present Value**
Cost Saving hosting on DigitalOcean	$ -	$ 4,73,504.00	$ 6,43,345.00	$ 7,55,045.00	$ 18,71,894.00	$ 15,29,425.00
Increased Productivity	$ -	$ 1,06,562.00	$ 1,21,645.00	$ 1,36,728.00	$ 3,64,935.00	$ 3,00,134.00
Cost Avoidance for dedicated DevOps/IT	$ -	$ -	$ 2,34,000.00	$ 4,68,000.00	$ 7,02,000.00	$ 5,45,004.00
Total Benifits(Risks-adjusted)	**$ -**	**$ 5,80,066.00**	**$ 9,98,990.00**	**$ 13,59,773.00**	**$ 29,38,829.00**	**$ 23,74,563.00**
Net Benefits	$ (40,425.00)	$ 3,51,741.00	$ 6,64,279.00	$ 9,54,586.00	$ 19,30,181.00	
ROI(%)	**191%**					
Payback Months	**<6**					

APPENDIX (10/15): Process of Modern Tech Stack Migration along with calculation of Investment Costs and Cost Savings of MDS

BAIN CAPABILITY NETWORK

Outdated Technology Stack

Current infrastructure limits agility and innovation for Zensoft, with legacy systems operating at high maintenance costs and reduced efficiency.

An outdated stack results in increased operational risks and hinders competitive advantage due to slow adaptation to market changes.

Selection of Modern Stack(MDS)

Identifying QT as the optimal modern tech stack provider for ZenSoft, offering scalability, security, and state-of-the-art technological capabilities.

The decision to partner with QT is driven by their proven track record in facilitating seamless integration with existing workflows and their robust support ecosystem.

Migration of existing stack to preferred Modern Stack

A comprehensive migration plan to QT's platform, ensuring minimal disruption to ongoing operations & strategic alignment with business goals.

The transition is governed by rigorous risk assessment and phased implementation, leveraging QT's expertise for a smooth & secure data transfer.

Enhanced Tech Stack with reduced downtime and lower costs

Post-migration, ZenSoft benefits from a significant reduction in system downtime, due to QT's resilient architecture and proactive monitoring systems.

The modernized stack results in lower total cost of ownership (TCO) & operational expenses, with measurable improvements in system performance.

INVESTMENT COST PROJECTIONS - TECH STACK MIGRATION

Associated Costs	2024E	2025E	2026E
Data Storage (Aggregates and stores data.)Sample vendors: PostgreSQL, MySQL, MongoDB, Amazon S3, Snowflake	$2,000	$10,000	$20,000
Data Processing / Compute (Operations run on data stored in a cloud data warehouse.)Sample vendors: dbt, Databricks, BigQuery	$20,000	$80,000	$150,000
ETL (Extract, Transform, Load) (Moves data from business systems into a data store and transforms it into a common format for reporting.)Sample vendors: Fivetran, Stitch, Apache Airflov	$12,000	$40,000	$75,000
Reverse ETL (Activates data in the data store by sending it back to business systems.)Sample vendors: Hightouch, Census	$8,000	$30,000	$50,000
Analytics (For creating reports and dashboards and performing ad-hoc analysis.)Sample vendors: Looker, Power BI, Tableau	$1,000	$6,000	$40,000
Observability / Catalog (Monitor and manage data and metadata across tools.)Sample vendors: Atlan, Monte Carlo, data.world, Informatica	$ -	$ -	$50,000
Technology Cost Subtotal	**$53,000**	**$166,000**	**$335,000**
Personnel (The humans required to build, manage, and maintain the various components of your data stack)Example titles: Data engineer, BI engineer, Analytics manager, Data analyst, Dat	$200,000	$600,000	$2,000,000
Technology + Personnel Costs Total	**$253,000**	**$766,000**	**$2,335,000**

APPENDIX (11/15): Comparison of Outdated and Modern stack highlights the need to shift, QT Group proves to be an ideal partner for this

BAIN CAPABILITY NETWORK

OUTDATED VS MODERN STACK		
Economic Factor	**Outdated Stack**	**Modern Stack**
Initial Investment	Lower upfront cost	Higher upfront cost
Maintenance Costs	Higher	Lower
Operational Efficiency	Lower (due to inefficiency in legacy hardware/software)	Higher (utilizes efficient, newer technologies)
Scalability	Costly and cumbersome	Cost-effective and flexible
Flexibility	Limited	High
Energy Consumption	Higher	Lower
System Downtime	More frequent due to maintenance and failures	Less frequent
Support and Vendor Availability	Decreasing as technology becomes obsolete	Increasing as technology is current
Operational Cost Reduction	-	Up to 30% reduction (Deloitte)
Initial Investment	Variable, often considered sunk cost	$10,000 to several million
Maintenance Costs	60-80% of IT budget (Gartner)	15-25% of IT budget post-modernization
Operational Efficiency	-	Up to 30% reduction in operational costs (Deloitte)
Scalability	Costly due to hardware/licenses	Savings over 35% for cloud services (IDC)
Flexibility	Limited, potentially resulting in lost opportunities	High, with faster integration of new services
Energy Consumption	Up to 100x more than modern facilities (DOE)	Up to 80% savings on energy costs
System Downtime	Up to $5,600 per minute (Gartner)	99.99% availability
Support and Vendor Availability	Decreasing as technology becomes obsolete	Increasing as technology is current
Operational Cost Reduction	-	Up to 30% reduction (Deloitte)

Hence, Modern Stack proves to be superior to primitive technology stack and there is a need to shift.

COMPETITOR ANALYSIS				
KPI	**Qt**	**Electron**	**WPF/UWP**	**Cocoa**
License Cost	Starting from $3,950/year per developer	Free (open-source)	Free with Visual Studio; Professional: $539/year, Enterprise: $2,999/year	Free with Xcode; Apple Developer Program: $99/year
Platform Support	Cross-platform (Windows, macOS, Linux, Android, iOS, embedded systems)	Cross-platform (mainly desktop platforms)	Windows (UWP for cross-Microsoft platforms)	macOS, iOS, tvOS, watchOS
Performance	High performance; optimized for embedded systems	Lower due to reliance on web technologies	High, native performance	High, native performance
Developer Community	Medium size with comprehensive documentation and support	Large, with extensive resources and community contributions	Large, supported by Microsoft's vast development ecosystem	Large, specifically within Apple's ecosystem
Additional Features	- Comprehensive C++ library classes - API for simplifying application development - High runtime performance with small	- Uses web technologies (HTML, JavaScript, CSS) - Good for web app porting	- Unified UI elements - Embedded in Windows OS	- Object-oriented API for UI development - Supports additional Apple platforms
Cost-Effectiveness	High initial cost but offers a broad range of features and cross-platform support	Cost-effective for leveraging web development skills for desktop applications	Costs associated with Visual Studio subscriptions but offers native performance	Free for development; costs associated with Apple ecosystem access

QT group identified as the most feasible partner.

APPENDIX (12/15): Investment Costs and Cost Savings of Modern Stack

OUTDATED STACK - INDUSTRY COST BREAKDOWN			
Economic Factor	Description	Final Cost	
Initial Investment	Costs for acquisition of legacy hardware and software platforms for desktop applications.	$ 15,000.00	*High due to scarcity and lack of support for legacy systems.*
Maintenance Costs	Ongoing technical support, updates, and servicing for outdated desktop software and hardware.	$ 27,500.00	*Increased due to rare parts and expertise required for older systems.*
Operational Efficiency	Increased man-hours and lower productivity due to older systems needing more manual intervention.	$ 17,900.00	*More manual work with outdated systems leads to higher labor costs.*
Scalability	Expense of adding resources or upgrading systems to meet new application requirements.	$ 10,000.00	*Costs for upgrades or new additions to handle growth with old systems.*
Flexibility	Cost implications of being unable to quickly adapt to new market demands or integrate with modern services.	$ 7,000.00	*Expenses related to adapting inflexible, outdated systems to new requirements.*
Energy Consumption	Higher operational costs due to energy-inefficient hardware typical of older desktop systems.	$ 8,300.00	*Inefficiency of old hardware leads to higher energy costs.*
System Downtime	Revenue loss and cost of business interruptions due to system instability and failures.	$ 3,400.00	*Losses from operational halts due to system instability.*
Support and Vendor Availability	Cost increase due to fewer vendors supporting outdated technology, leading to premium pricing on necessary parts.	$ 8,790.00	*Fewer service providers leads to premium prices for support.*
Operational Cost Reduction	Lack of cost reduction opportunities with an outdated stack. No number is assigned here as it's not applicable.	$ 2,000.00	*Limited by outdated tech, resulting in minimal savings.*
Total		**$ 99,890.00**	

MODERN STACK - INDUSTRY COST BREAKDOWN			
Associated Costs	Description	Final Cost	
Qt Licensing	Qt for Application Development (Enterprise)	$ 21,300.00	*5 developers per year needed for a typical enterprise to run MDS*
Development Hardware	Workstations for Developers	$ 12,500.00	*Each developer requires a high-performance workstation priced at $2,500.*
Software & Tools	Additional Software Licenses	$ 2,500.00	*Additional software includes graphics and testing tools necessary for development.*
Training and Support	Qt Training for Team	$ 5,000.00	*One-time training session for the team on Qt usage and an annual premium support subscripti*
	Premium Support Package	$ 10,000.00	
Cloud Services	Cloud-Based Dev and Testing Environment	$ 6,000.00	*Monthly cost for cloud-based development and testing environments.*
Miscellaneous Costs	Third-party Integrations and Libraries	$ 3,000.00	
	Contingency Fund	$ 5,000.00	*Includes clouds for integration third-party libararies, a contingency fund for unforeseen expens*
	Marketing and Launch Activities	$ 10,000.00	
Total		$ 75,300.00	

COST SAVINGS CALCULATION

Total Cost from Outdated Stack	99890.00
Total Cost from Modern Stack	75300.00
Cost Savings	**24590.00**
Cost Savings(%)	**24.62%**

APPENDIX (13/15): Cost Savings through enhancing AI Capabilities

ROI Analysis Model - DigitalOcean and CTO AI						
Service	**Provider**	**Cost Description**	**Estimated Monthly Cost Before**	**Estimated Monthly Cost After**	**Monthly Savings**	**Source**
Basic Cloud Services	Digital Ocean	Basic infrastructure costs including servers, storage, and network	$ 10,000.00	$ 7,000.00	$ 3,000.00	Digital Ocean Pricing
Advanced DevOps Services	CTO.ai	Advanced DevOps automation and management services	$ 3,500.00	$ 2,450.00	$ 1,050.00	CTO.ai Pricing
AI Framework Installation	Digital Ocean	Cost for setting up AI frameworks like TensorFlow on cloud servers	$ 2,000.00	$ 1,000.00	$ 1,000.00	Digital Ocean Solutions
Kubernetes Management	CTO.ai	Managed Kubernetes services to streamline deployment and operations	$ 4,000.00	$ 2,800.00	$ 1,200.00	CTO.ai Kubernetes Solutions
Data Management	Digital Ocean	Costs associated with large-scale data handling and storage solutions	$ 5,000.00	$ 3,500.00	$ 1,500.00	Digital Ocean Data Management
Total(Monthly)			**$ 24,500.00**	**$ 16,750.00**	**$ 7,750.00**	
Total(Annual)			**$ 2,94,000.00**	**$ 2,01,000.00**	**$ 93,000.00**	
Annual Savings(%)	**31.63%**					

Integration of AI Capabilities through existing partnership with CTO.ai leads to an aggregate cost savings of 31.63%, driving profitability.

APPENDIX (14/15): Lean Training for internal personnel – Examining two scenarios for its costing

Zensoft grapples with hefty IT costs due to its reliance on physical servers. Two possible solutions emerge: downsizing part of its IT staff and hiring cloud experts for a swift transition, or gradually reducing the team while internally training staff on cloud engineering, supplementing with new cloud engineers as necessary. Each strategy carries distinct implications for Zensoft's operations and workforce.

PARTICULARS	XXX	ASSUMPTIONS
Average IT Employee Salary at Zensoft ($)	65000	*Based on Industry Averages prevalent in US*
Annual IT Staffing Costs for Zensoft ($)	1000000	*Information provided in the case*
Average No. of employees	15	
Case 1: Suppose 40% of the Current IT Staff is Layed-off and Cloud Experts are Hired	**XXX**	
Severance Cost 40% IT Staff due to migration to Cloud	682500	*Severance cost includes 1.5 months of initial salary*
Salary of Cloud Experts Hired (3-4 depending upon the size)	182000	*~ 30% cost savings seen in cloud professionals as compared to physical technicians*
TOTAL COST INVOLVED IN CASE 1	**864500**	
Case 2: Leaner IT Staff undergoes Cloud Training and Required Cloud Experts are Hired	**XXX**	
Severance Cost of 15 % It Staff due to migration to Cloud	195000	*Severance cost includes 1.5 months of initial Salary*
Training Costs (Courses + Certifications + Practical Learning Simulations)	320000	*Assuming a 60:40 split between setup and training costs as incurred by zensoft under DigitalOcean's Package*
Salary of Cloud Experts Hired (2-3 Required Individuals)	136500	*~ 30% cost savings seen in cloud professionals as compared to physical technicians*
TOTAL COST INVOLVED IN CASE 2	**651500**	
POTENTIAL ADDITIONAL COSTS INVOLVED IN CASE 1 (%)	**25**	*This excludes potential losses due to reputational damage of laying-off with zensoft being the best place to work*

Hence, Case 2 seems more advantageous and should be chosen by Zensoft to tackle the challenges encountered during the hybrid cloud migration.

APPENDIX (15/15): Computation of ESG Score, Pre and Post Solutions

ESG Scorecard Pillar	Zensoft Issue	Point	%Weight	Weighted Score
Environment			**0.3**	
	Energy Management	1	0.075	0.075
	Greenhouse Gas Emissions	1	0.15	0.15
	Waste Management	2	0.075	0.15
Social			**0.35**	
	Labour Management	2	0.1375	0.275
	Diversity, Equity & Inclusion	2	0.1375	0.275
	Community Engagement	2	0.075	0.15
Governance			**0.35**	
	Data Privacy	1	0.15	0.15
	Corporate Transparency & Disclosure	2	0.1	0.2
	Management Structure & Compensation	2	0.1	0.2

ESG Scorecard Pillar	Zensoft Issue	Point	%Weight	Weighted Score
Environment			**0.3**	
	Energy Management	2	0.075	0.15
	Greenhouse Gas Emissions	3	0.15	0.45
	Waste Management	1	0.075	0.075
Social			**0.35**	
	Labour Management	3	0.1375	0.4125
	Diversity, Equity & Inclusion	2	0.1375	0.275
	Community Engagement	2	0.075	0.15
Governance			**0.35**	
	Data Privacy	3	0.15	0.45
	Corporate Transparency & Disclosure	2	0.1	0.2
	Management Structure & Compensation	2	0.1	0.2

Total Score 1.625/3

Total Score 2.363/3

Assumptions

ESG Score Computation

- ✓ The table represents an excerpt of the Refinitv ESG scorecard and highlights only key components necessary for this case.
- ✓ The distribution of the overall weights for E (30%), S (35%) and G (35%) is the average weightage based on Industry Median values prevalent in IT & Software segment.
- ✓ The distribution across various factors present in each of Environment, Social & Governance varies owing to steps taken in that direction.
- ✓ The shifts from 1 to 3 value points is dependent on extent of change or step taken.


THANK YOU!